

EE Dept, IIT Bombay
Academic Year: 2025-2026, Semester I (Autumn)

Course: MS101 Makerspace

EE Lecture 10

Transistors

Topics

1. Transistor Basics
2. Semiconductor Fundamentals
3. Bipolar Junction Transistor (BJT)
4. Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)
5. Switches Using BJTs & MOSFETs
6. Digital Logic Gates Using MOSFETs

Reference: AS Sedra, KC Smith, TC Carusone, & V Gaudet, Microelectronic Circuits, 8th ed., Oxford University Press, 2020. Chs. 1, 2, 11, 13, 15.

1. Transistor Basics

1.1 Transistor

A transistor is a semiconductor device with three terminals, used in analog & digital applications. It is mostly used as a 'two-port device' with one terminal common between the input and output ports.

Commonly used transistors

- Bipolar Junction Transistor (BJT)
- Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)

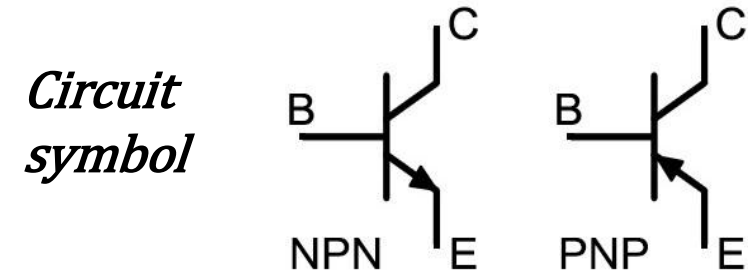
Both types are used in integrated circuits (ICs) and are also available as discrete devices.

Main Applications: amplifiers, oscillators, loads, switches, digital circuits.

1.2 Bipolar Junction Transistor (BJT): symbol & model

Types: NPN, PNP

Terminals: Emitter (E), Base (B), Collector (C)



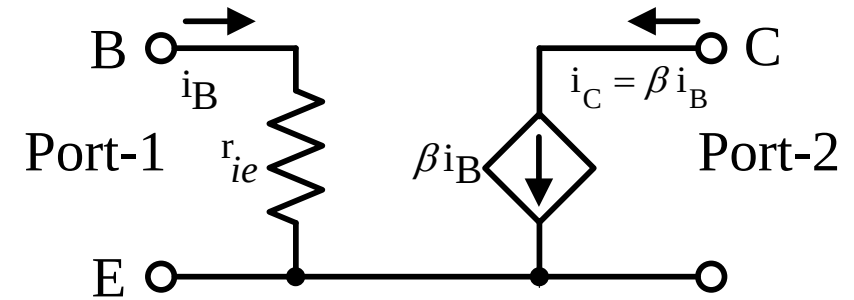
Two-port configurations with one terminal common between the input & output ports

- Common emitter (CE)
- Common base (CB)
- Common collector (CC)

Simplified small-signal model for CE configuration

(current-controlled current source)

- Input: base current i_B
- Output: collector current $i_C = \beta i_B$
- β : Current gain
- r_{ie} : base-emitter input resistance



1.3 Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET): symbol & model

Types

- N-channel enhancement mode
- P-channel enhancement mode
- N-channel depletion mode
- P-channel depletion mode

Terminals: Source (S), Gate (G), Drain (D), Substrate / body (B)

Two-port configurations with one terminal common between the input & output ports

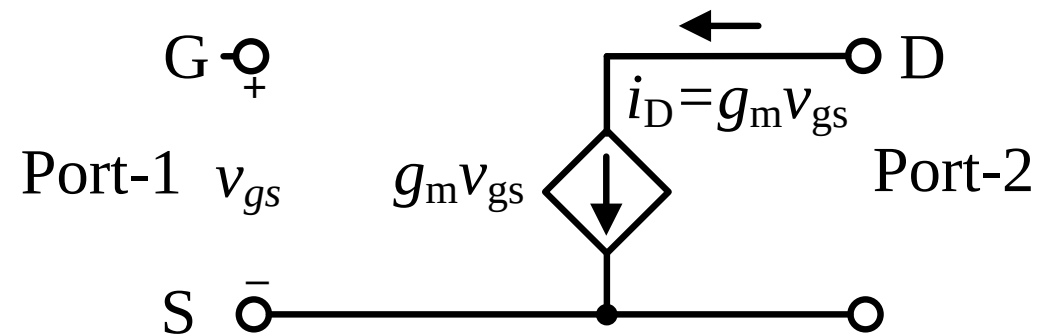
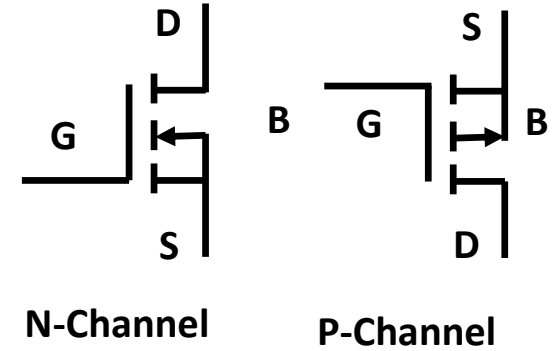
- Common source (CS)
- Common gate (CG)
- Common drain (CD)

Simplified small-signal model for CS configuration

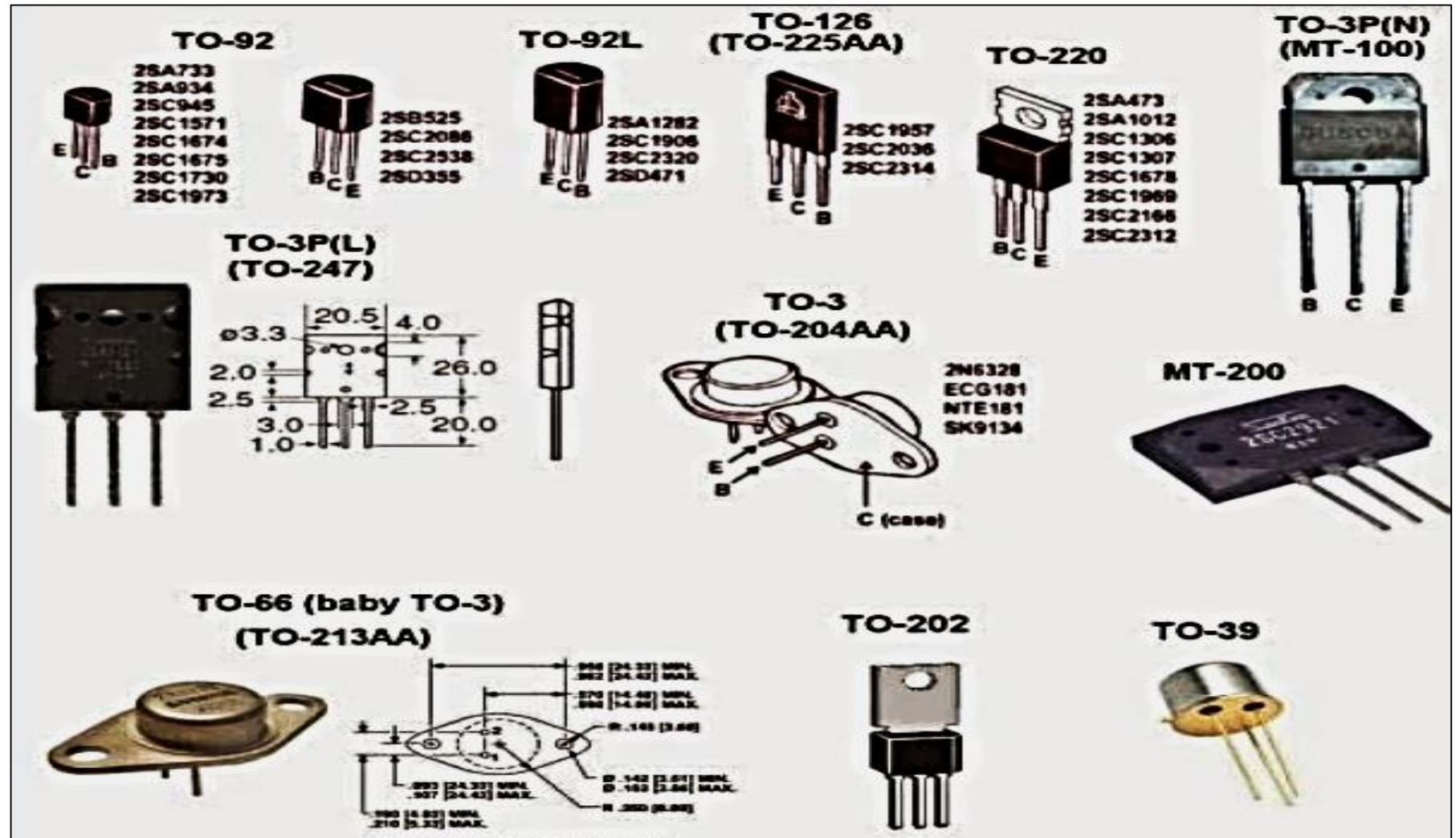
(voltage-controlled current source)

- Input: gate-source voltage v_{GS}
- Output: drain current $i_D = g_m v_{GS}$
- g_m : trans-conductance
- $i_G \approx 0 \Rightarrow$ Almost no power from the input source.

*Circuit symbol
(enhancement
mode) with
B-S shorted*

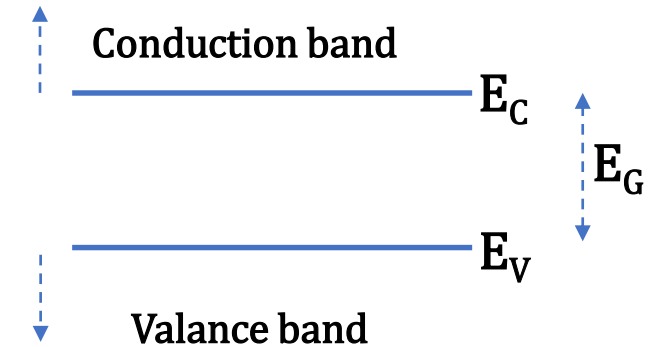


1.4 Some Commercially Available Discrete Transistor Packages

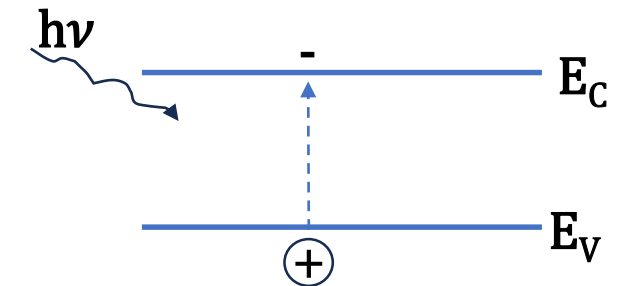


2. Semiconductor fundamentals (1) - Plus 2 Physics refresher

- From an electronic perspective, solids can be classified into 3 categories based on their band gap.
- The most generic category is the semiconductor.
- In a semiconductor, the electrons are distributed in bands of energy states called the valence band and the conduction band.
- The two bands are separated by a forbidden gap, where there are no energy states.
- The separation between the top edge of the valence band (E_V) and the bottom edge of the conduction band (E_C) is called the bandgap, E_G . For silicon, the most commonly used semiconductor, $E_G = 1.1$ eV
- At absolute zero temperature, the valence band of a semiconductor is completely filled with electrons, and the conduction band is empty.
- At absolute zero temperature, no net electron flow is possible within the valence band in response to an applied electric field as no empty energy states are available. Since there are no electrons in the conduction band, there won't be an electron flow in the conduction band either.
- Electrons can be excited from the valence band to the conduction band by providing them with energy by heating the semiconductor or shining light with photon energy $\geq E_G$.



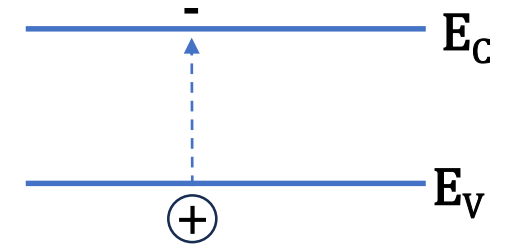
(1) band diagram of a semiconductor



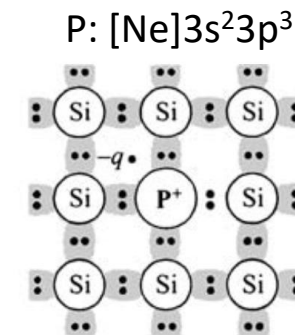
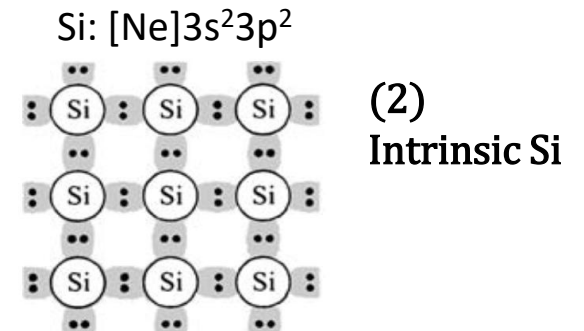
(2) Electron – hole pair generation due to optical excitation

2. Semiconductor fundamentals (2)

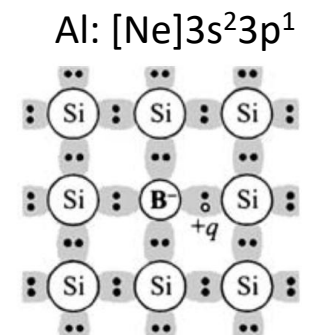
- Excitation of an electron to the conduction band leaves an empty energy state in the valence band. This empty state is called a hole.
- The hole can be treated as a positively charged particle.
- The process of creating electron-hole pairs described above is called generation.
- The excited electrons can relax back to the empty state (hole). This is called recombination.
- The concentration of conduction band electrons can be increased by doping silicon with pentavalent elements like phosphorus and arsenic.
- Si doped with P or As is an n-type silicon. In n-type Si, the concentration of conduction band electrons is higher than the concentration of holes. Hence, in n-type Si, electrons are the majority carriers and holes are the minority carriers.
- The concentration of holes can be increased by doping silicon with trivalent elements like boron and gallium, to produce p-type Si. In p-type Si, holes are the majority carriers and electrons are the minority carriers in n-type Si.



(1) Electron – hole pair
recombination



(3) n-type Si



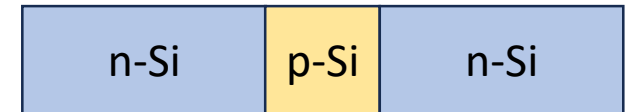
(4) p-type Si

2. Semiconductor fundamentals (3)

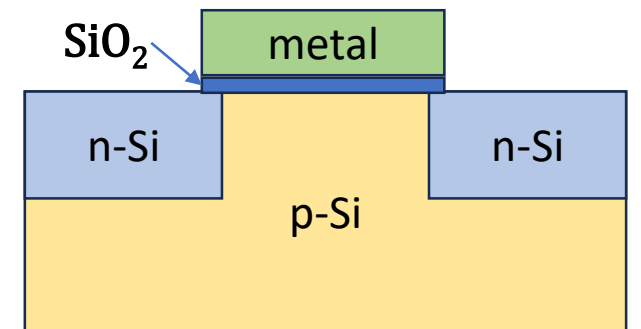
- A pn junction diode consists of two adjacent regions in the semiconductor, one of which is n-type doped and the other p-type doped.
- Under forward bias to the pn junction (see lecture), holes diffuse from the p-side to the n-side, and electrons from the n-side to the p-side, leading to a current from the p-side to the n-side.
- More complex devices like bipolar junction transistors (BJT) can be made by adding more n- or p-regions in Si as shown.
- A metal-oxide-semiconductor field effect transistor (MOSFET) can be made using a BJT like structure, and placing an oxide (typically SiO₂) and metal (typically aluminium) on Si as shown.



(1) p-n junction diode



(2) npn BJT



(3) n-channel MOSFET

3. Bipolar Junction Transistor (BJT)

3.1. Bipolar Junction Transistors

Semiconductor device with three doped regions, each with a terminal.

Types: NPN, PNP

Terminals: Emitter (E), Base (B), Collector (C)

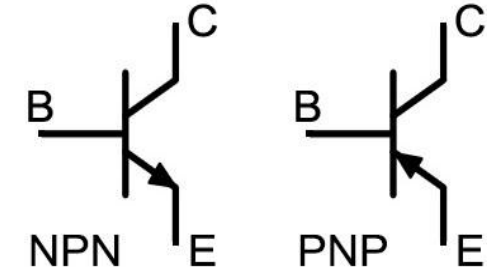
NPN transistor

Emitter (E): heavily doped N region

Base (B): very thin P region

Collector (C): lightly doped N region

*Circuit
symbol*



PNP transistor

Emitter (E): heavily doped P region

Base (B): very thin N region

Collector (C): lightly doped P region

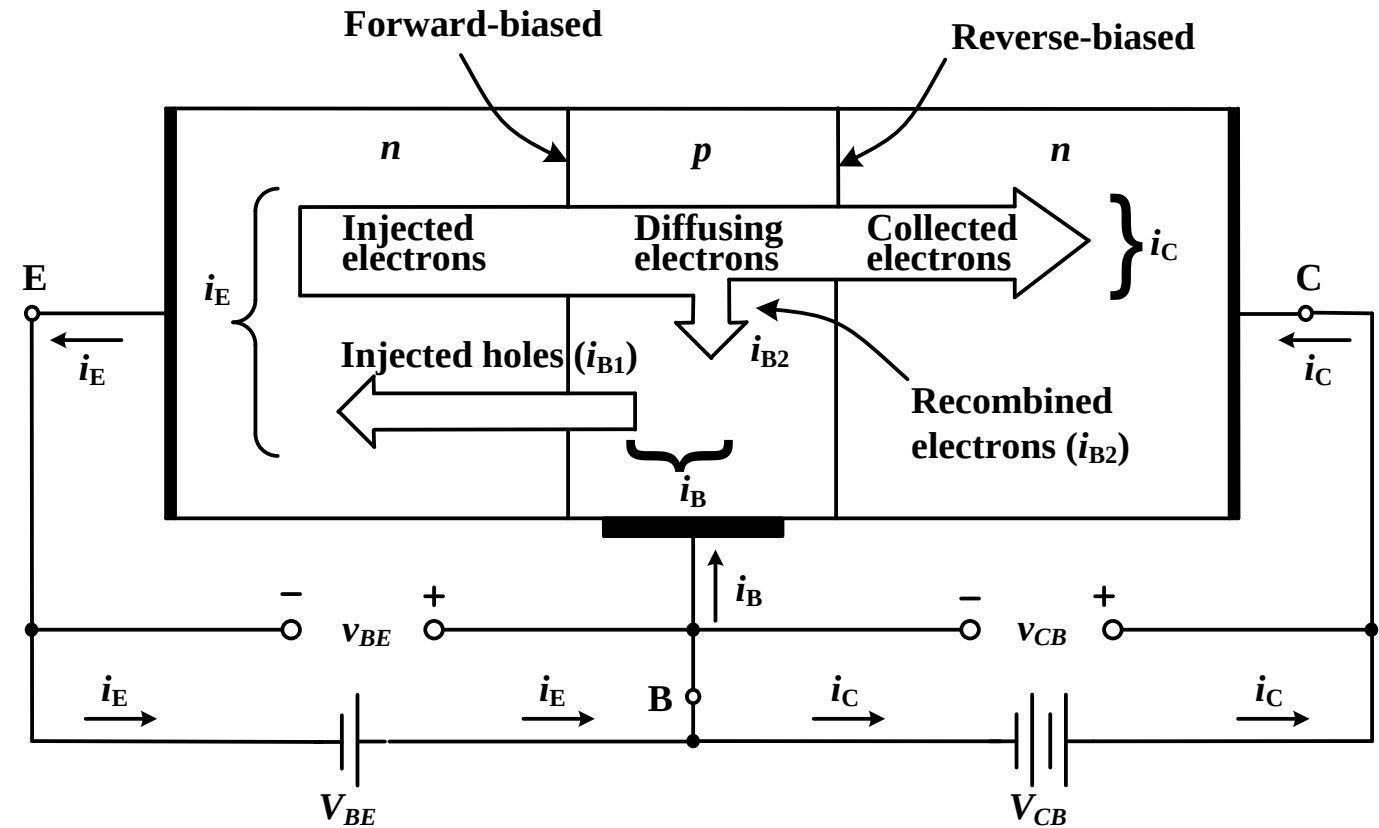
3.2 NPN Transistor Operation

Majority carriers in the base (P) region: holes.

Minority carriers in the base region: electrons.

Flow of majority and minority carriers controlled by the voltages across the BE and BC junctions.

Three useful operation modes: active, saturation, cut-off.



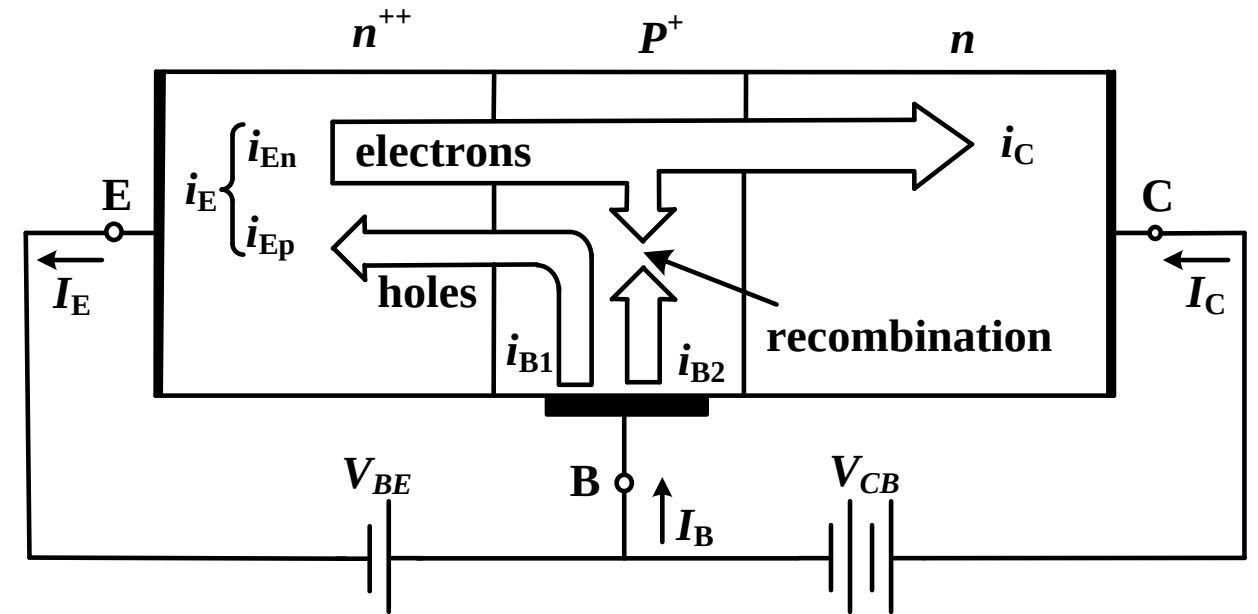
Active mode: BE junction is forward biased. BC junction is reversed biased.

Saturation mode: BE & BC junctions are forward biased.

Cut-off mode: BE & BC junctions are reverse biased.

Active mode operation

- The emitter – base junction is forward biased. Electrons diffuse from emitter to base
- About 1-2% of these electrons will recombine with the holes in the base, leading to a base current.
- The remaining electrons are collected by the collector.



Current relations

KCL: $I_E = I_B + I_C$

Transistor action

$$\Delta I_C = \alpha \Delta I_E \quad (\alpha \approx 0.90 - 0.99)$$

$$\Delta I_C = \beta \Delta I_B$$

$$I_C \approx \beta I_B \quad (\text{approximate relation})$$

Relation between α & β

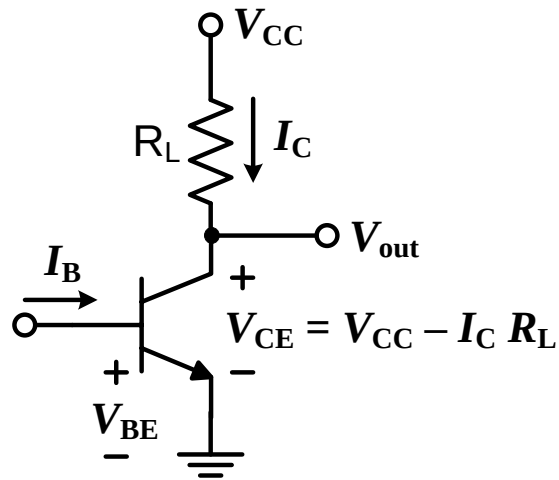
$$\Delta I_E = \Delta I_B + \Delta I_C \Rightarrow \Delta I_B = \Delta I_E - \Delta I_C = (1/\alpha - 1) \Delta I_C$$

$$\Rightarrow \Delta I_C = [\alpha/(1-\alpha)] \Delta I_B = \beta \Delta I_B$$

$$\Rightarrow \beta = \alpha/(1-\alpha)$$

$$\text{Example: } \alpha = 0.98 \Rightarrow \beta = 0.98/0.02 = 49$$

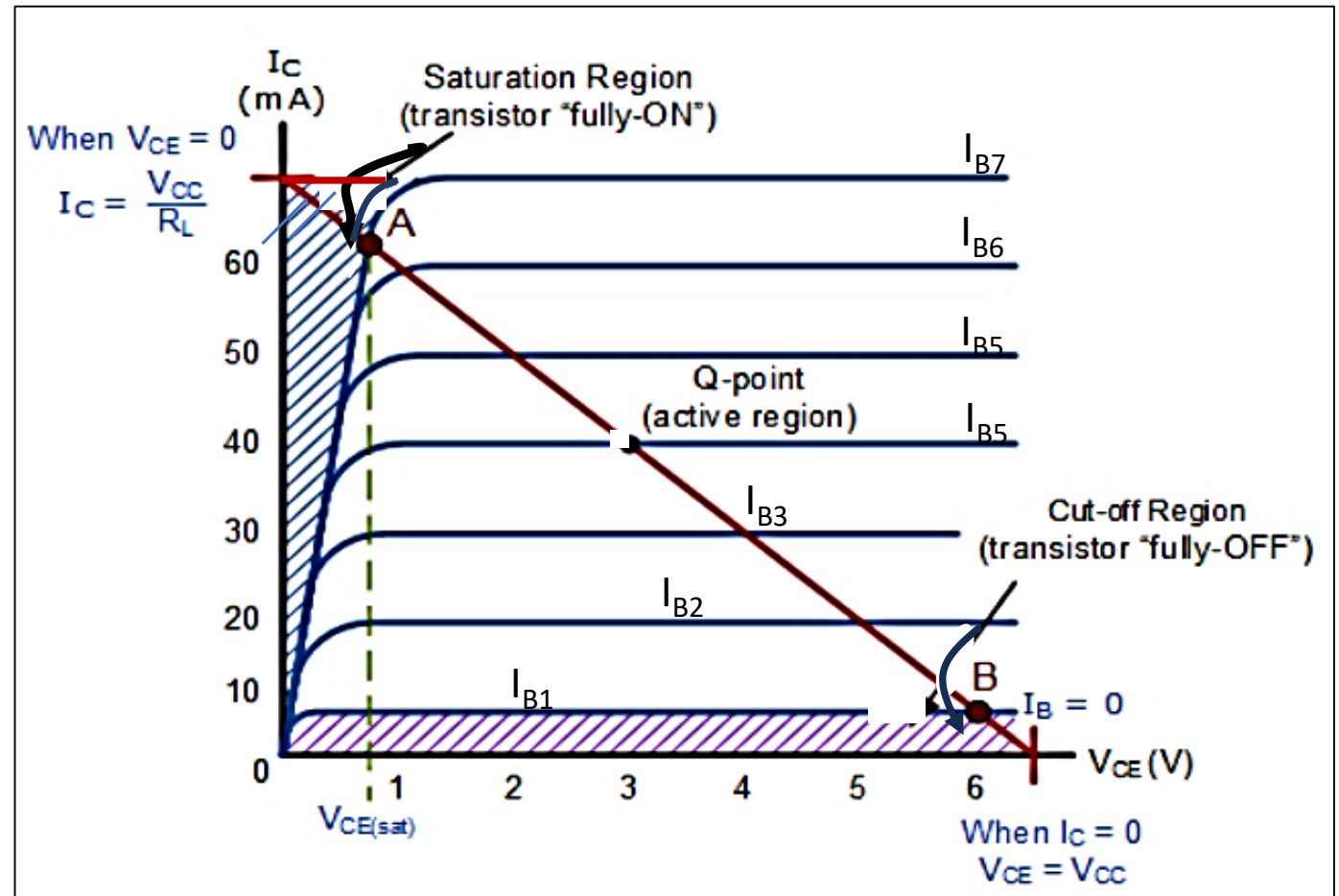
3.3 NPN Transistor Characteristics: I_C vs V_{CE}



Operation modes

- Active: Forward biased BE junction, reverse biased BC junction.
- ON (saturation): Forward biased BE & BC junctions. High I_B , high I_C , low V_{CE} .
- OFF (cut-off): Reverse biased BE & BC junctions. $I_B \approx 0$, $I_C \approx 0$, High V_{CE} .

Typical ' $V_{CE} - I_C$ ' characteristics ($I_{B1} < I_{B2} < I_{B3} < I_{B4} < I_{B5} < I_{B6} < I_{B7}$). Operating points for switching action: A (on) & B (off).



3.4 NPN Transistor as a Switch

Device operation in ON (saturation) and OFF (cut-off) modes.

- Common emitter (CE) configuration
- Control input on the base side
- Load on the collector side
- ON / OFF control by changing the base current I_B

OFF: $V_{BE} < V_\gamma$

BE junction cutin voltage: $V_\gamma \approx 0.5 \text{ V}$

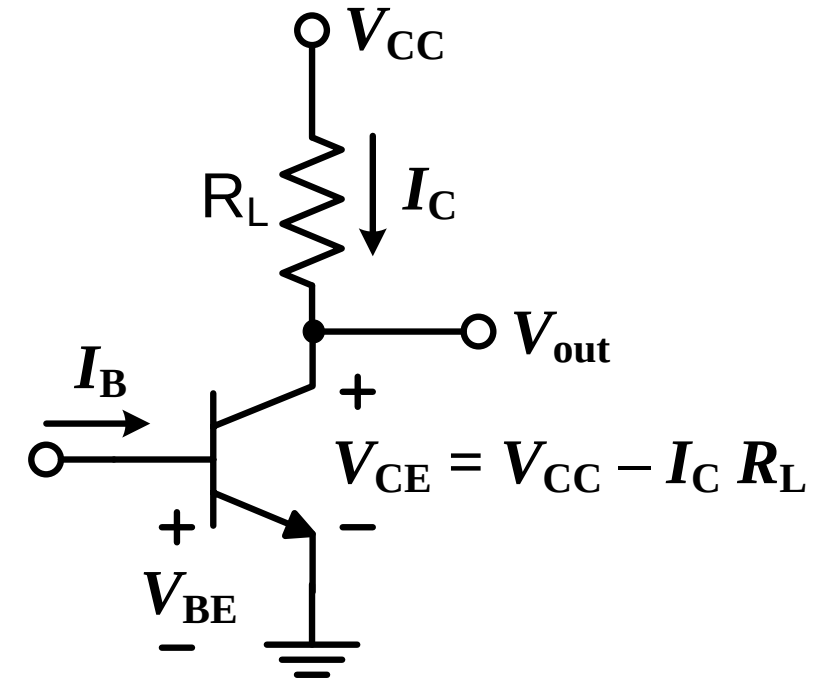
$I_B \approx 0.$ $I_C \approx 0.$ $V_{CE} \approx V_{CC}.$ $V_{BC} < 0.$

Heavily ON: $I_B > I_C / \beta$

$V_{BE} = V_{BES},$ $V_{CE} = V_{CES}$

Saturation voltages: $V_{BES} \approx 0.8 \text{ V},$ $V_{CES} \approx 0.2 \text{ V})$

$I_C = (V_{CC} - V_{CES}) / R_L.$ $V_{BC} > 0.$



4. Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)

4.1 MOSFET

Terminals: Source (S), Drain (D), Gate (G), Substrate / Body (B)

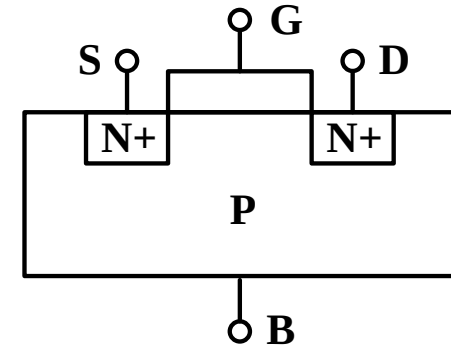
Structure

- Doped semiconductor region with substrate or body (B) terminal with two heavily doped regions as source (S) and drain (D) terminals.
- Thin oxide layer insulation
- Conductor (metal) layer with gate (G) terminal

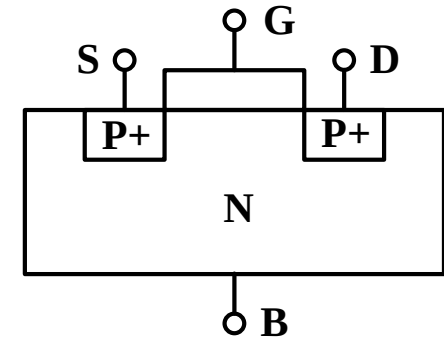
Operation:

- Flow of majority carriers in the D-S channel is controlled by the gate-channel voltage.
- Voltage-controlled current source device.
- Gate insulated from the channel $\Rightarrow$ Nearly zero gate current (I_G)
 $\Rightarrow$ Almost no power taken from the input source.

N-channel enhancement-mode MOSFET



P-channel enhancement-mode MOSFET



4.2 MOSFET Types & Symbols

Types

- N-channel enhancement-mode
- P-channel enhancement-mode
- N-channel depletion-mode
- P-channel depletion-mode

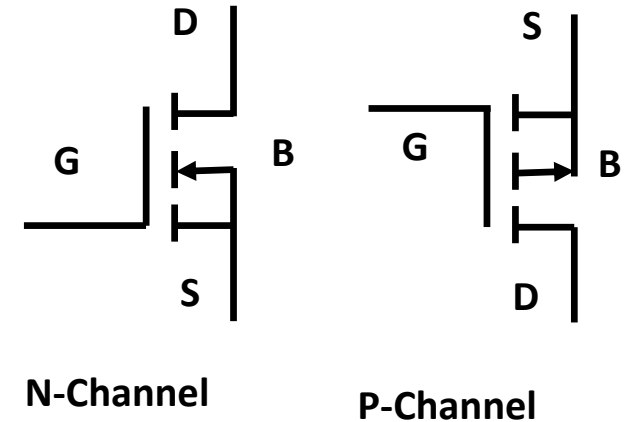
Terminals: Source (S), Gate (G), Drain (D), Substrate / Body (B)

Commonly used MOSFETs

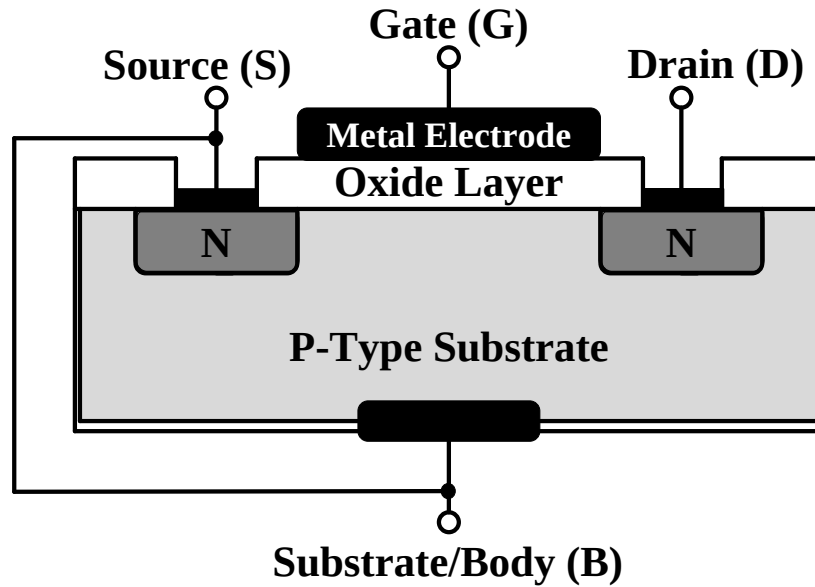
N-channel & P-channel enhancement-mode

Enhancement-mode device symbols

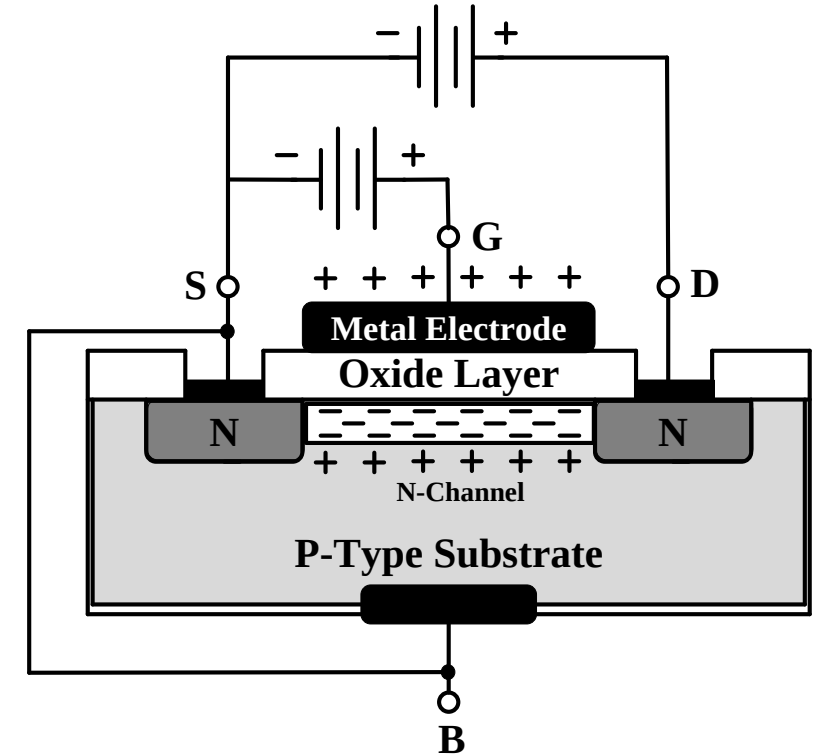
(B-S usually shorted in switching & logic circuits)



4.3 N-channel Enhancement-Mode MOSFET Operation



- The D & S terminals are interchangeable. $V_{DS} > 0$.
- No D-S channel is available at $V_{GS} = 0$.
- For N-channel, the D-S channel is formed for $V_{GS} > V_T$ (threshold voltage > 0).
- OFF-state: $V_{GS} < V_T$. Very high resistance for drain-source current flow.
- ON-state: $V_{GS} > V_T + \text{a few V}$. Low resistance for drain-source current flow.



MOSFET Type

N-channel
Enhancement

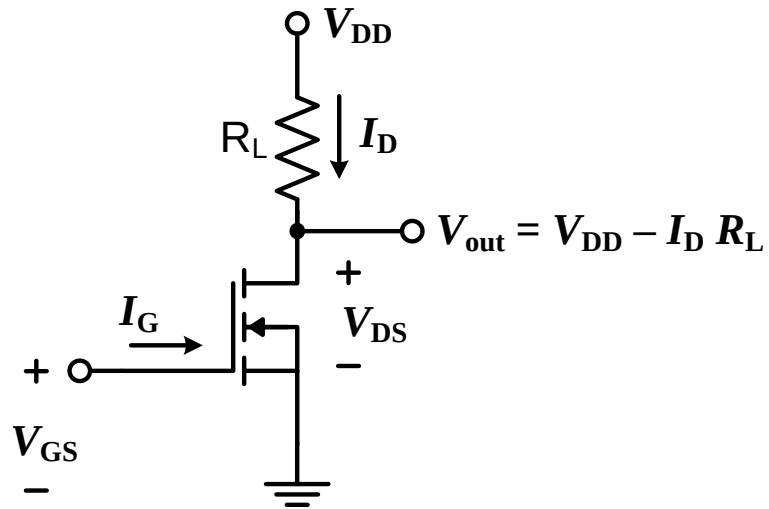
P-channel
Enhancement

Condition for Switching

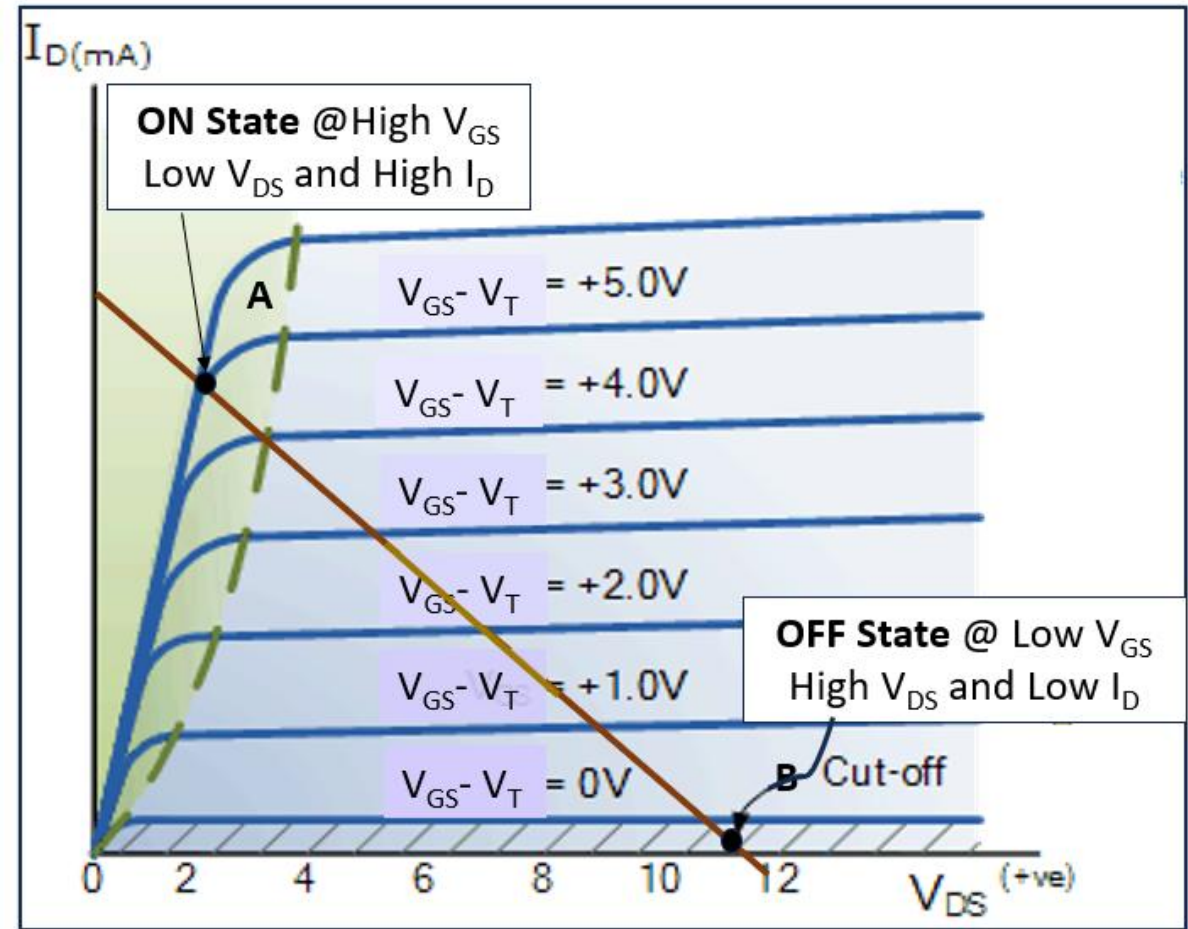
OFF for $V_{GS} < V_T$ (0.6 to 1 V)

OFF for $V_{GS} > V_T$ (-1.0 to -0.6 V)

4.4 N-channel Enhancement-Mode MOSFET Characteristics: I_D vs V_{DS}



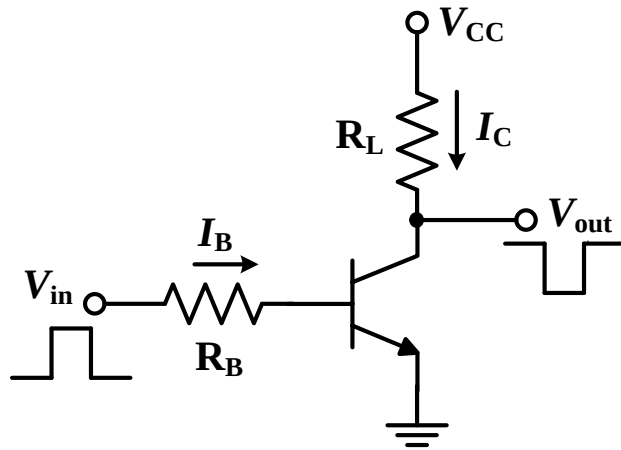
- $V_{DS} > 0$. D-S conduction starts when the N-channel is formed between the D & S terminals.
- V_T is the minimum V_{GS} for channel formation. Channel is formed for $V_{GS} > V_T$. Channel resistance decreases as V_{GS} increases further.
- OFF state: $V_{GS} < V_T$, $I_G \approx 0$, $I_D \approx 0$, $V_{DS} \approx V_{DD}$
- ON state: $V_{GS} > V_T + \text{a few V}$.
 $I_G \approx 0$, $V_{DS} \approx 0$, $I_D \approx V_{DD} / R_L$



Typical ' V_{DS} - I_D ' characteristics for N channel Enhancement mode MOSFET
 A & B: operating points for switching action

5. Switches Using Transistors

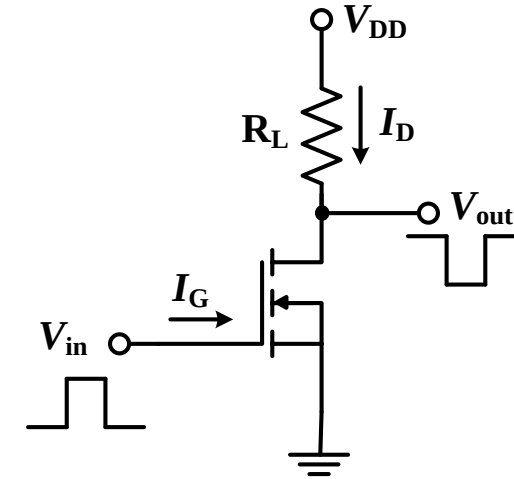
5.1 BJT Switch



BJT switch for load R_L connected to +ve supply end (V_{CC})

- Control input: V_{in}
- NPN BJT in common-emitter (CE) configuration
- R_B for limiting I_B
- OFF: $I_B \approx 0$ ($V_{in} = \text{low}$)
- ON: $I_B > I_C/\beta$ ($V_{in} = \text{high}$)

5.2 MOSFET Switch

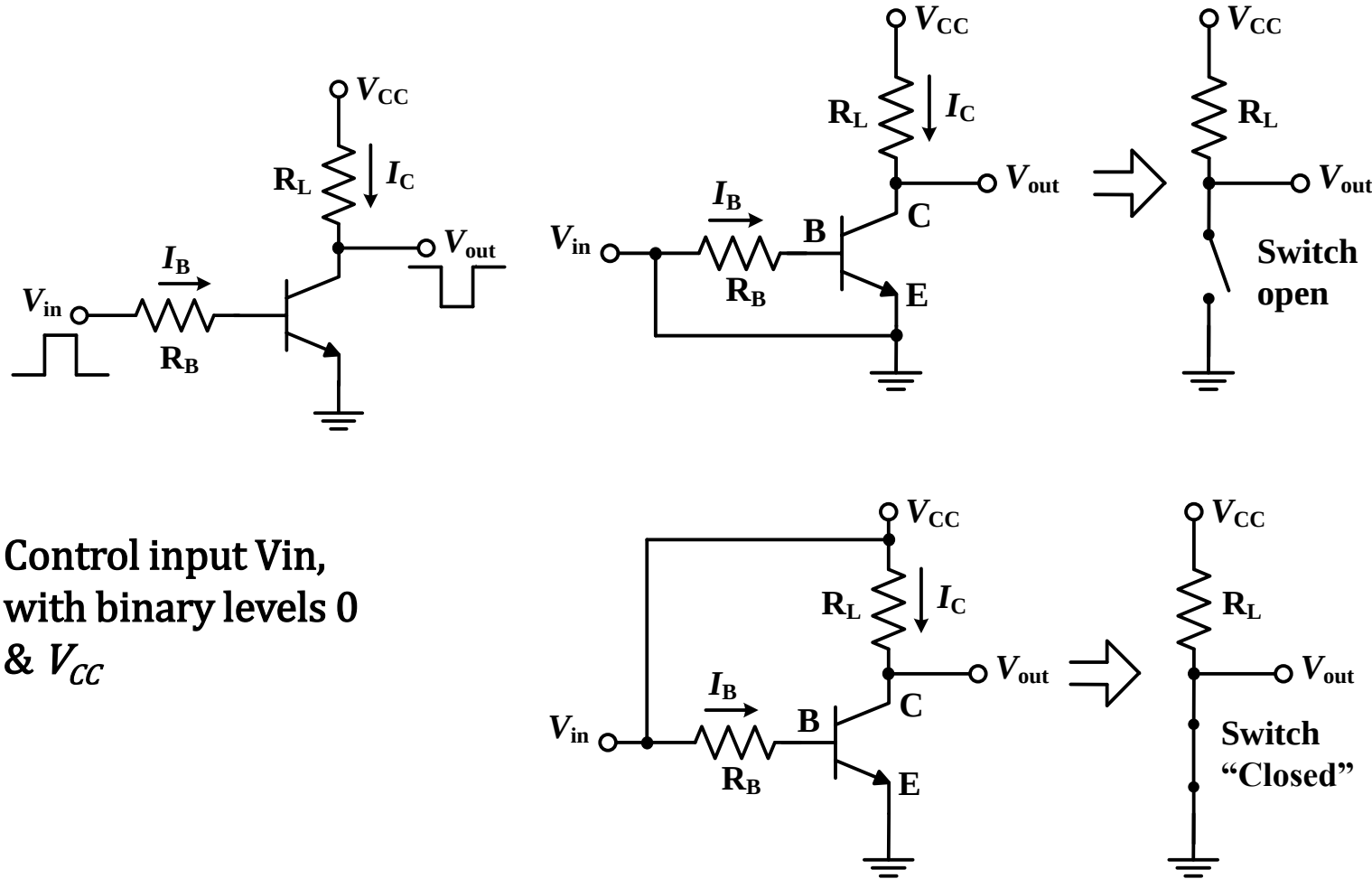


MOSFET switch for load connected to +ve supply end (V_{DD})

- Control input: V_{in}
- N-channel MOSFET in common-source (CS) configuration
- OFF: $V_{in} < V_T$ ($V_{in} = \text{low}$)
- ON: $V_{in} > V_T + \text{a few V}$ ($V_{in} = \text{high}$)
- $I_G \approx 0$ in both conditions

Load R_L connected to -ve supply end (Ground): Use PNP BJT in CE configuration or P-channel MOSFET in CS configuration.

5.3 NPN BJT Switch Operation



Control input V_{in} ,
with binary levels 0
& V_{CC}

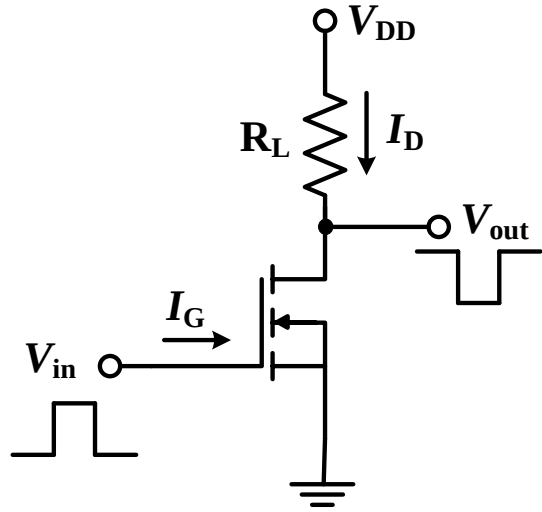
(a) Switch open (transistor in OFF or cut-off state mode): $V_{IN} = 0$

- $V_{BE} = 0 < 0.5V$.
- BJT operates as open switch.
 $I_B \approx 0$. $V_{CE} \approx V_{CC}$. $I_C \approx 0$.
- BE junction is not forward biased.
- BC junction is reverse biased.

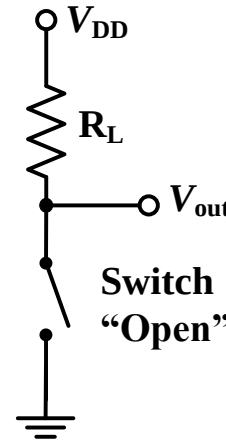
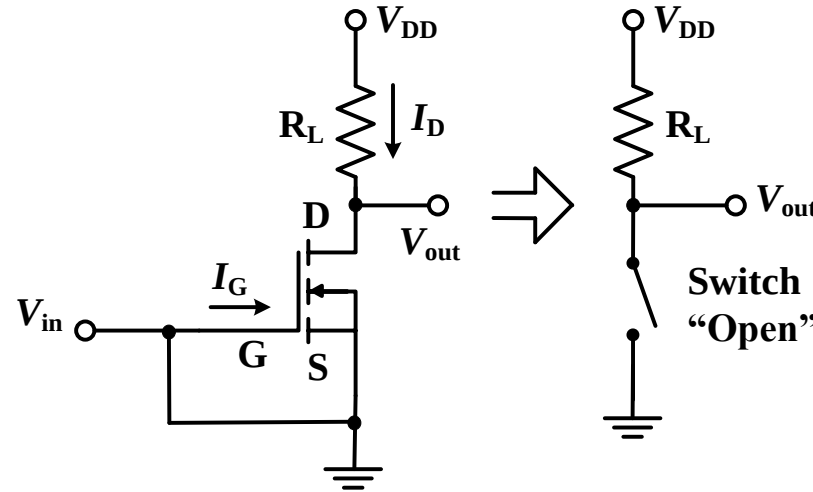
(b) Switch closed (transistor in ON or saturation mode): $V_{IN} = V_{CC}$

- BE junction is forward biased. I_B is limited by R_B . V_{BE} reaches saturation value $V_{BES} (\approx 0.8V)$.
- Collector voltage drops until BC junction gets forward biased. V_{CE} reaches saturation value $V_{CES} (\approx 0.2V)$.
- BJT operates as closed switch.
- $V_{BE} = 0.8V$. $V_{out} = V_{CES} = 0.2V$.
 $I_B = (V_{CC} - 0.8) / R_B$.
 $I_C = (V_{CC} - V_{CES}) / R_L$.

5.4 N-channel MOSFET Switch Operation



Control input V_{in} with binary levels 0 & V_{DD}



Let $V_{DD} = 12\text{ V}$, $V_T = 2\text{ V}$,

(a) Switch open: $V_{in} = 0$

$$V_{GS} = 0 < V_T$$

$$I_G \approx 0$$

$$V_{DS} \approx V_{DD}$$

$$I_D \approx 0.$$

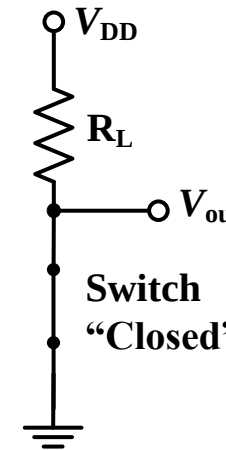
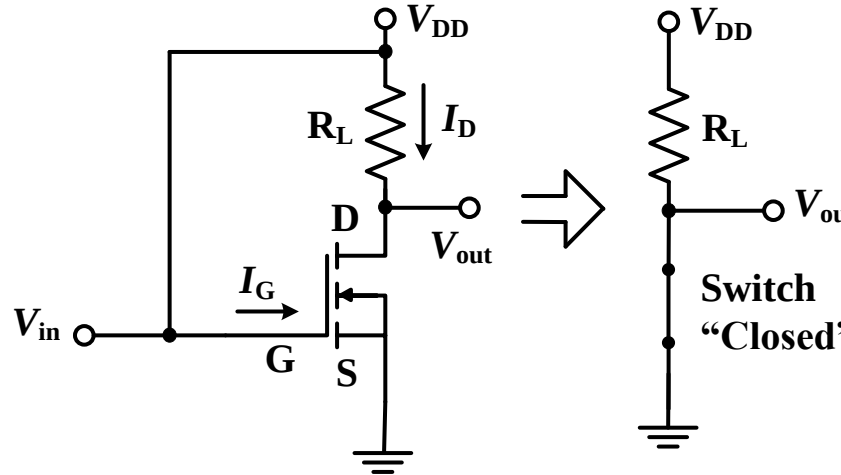
(b) Switch closed: $V_{in} = V_{DD}$

$$V_{GS} = 12\text{ V} > V_T$$

$$I_G \approx 0$$

$$V_{DS} \approx 0$$

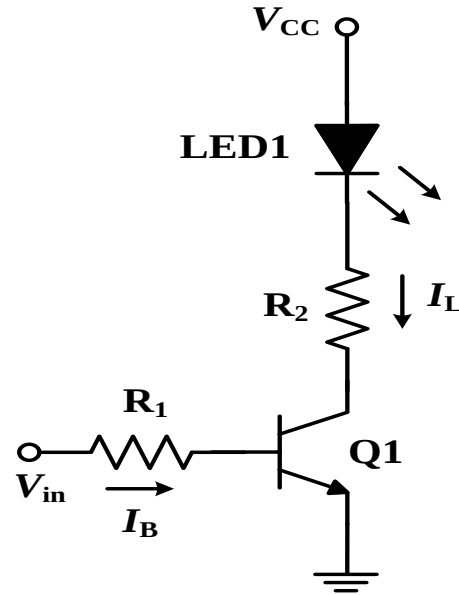
$$I_D = V_{DD}/R_L$$



5.5 NPN and PNP BJT Switches for LED Control

NPN switch for load connected to +ve supply end (V_{cc})

Given: $V_{cc} = 5\text{ V}$.
 V_{in} (LED off) = 0 V.
 V_{in} (LED on) = 5 V.
 $\beta > 50$.
 I_L (full brightness) = 10 mA.
LED voltage drop = 2 V.

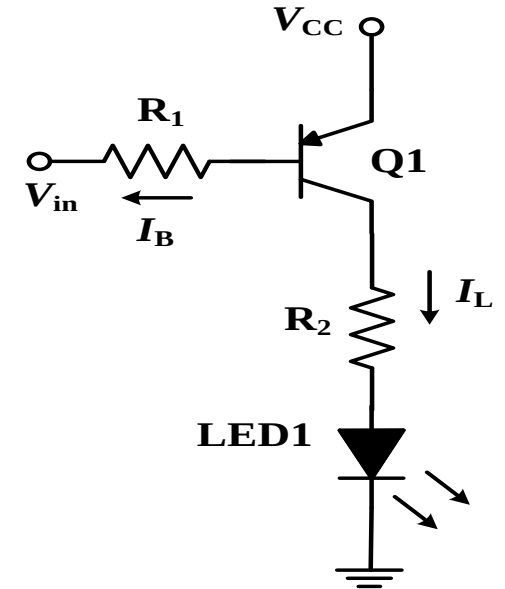


Solution

$I_L = (V_{CC} - V_{LED} - V_{CES}) / R_2 = (5 - 2 - 0.2) / R_2 > 10\text{ mA}$
 $\Rightarrow R_2 < (2.8\text{ V}) / (10\text{ mA}) = 280\ \Omega$.
Select $R_2 = 270\ \Omega$. $\Rightarrow I_L = 10.3\text{ mA}$.
 $I_B = [(V_{in})_{\text{high}} - V_{BES} - 0] / R_1 > I_L / \beta_{\text{min}}$
 $\Rightarrow (5 - 0.8 - 0) / R_1 > 10.3 / 50 \Rightarrow R_1 < 20.38\text{ k}\Omega$.
Select $R_1 = 18\text{ k}\Omega$.

PNP switch for load connected to -ve supply end (GND)

Given: $V_{cc} = 5\text{ V}$.
 V_{in} (LED off) = 5 V.
 V_{in} (LED on) = 0 V.
 $\beta > 50$.
 I_L (full brightness) = 10 mA.
LED voltage drop = 2 V.



Solution

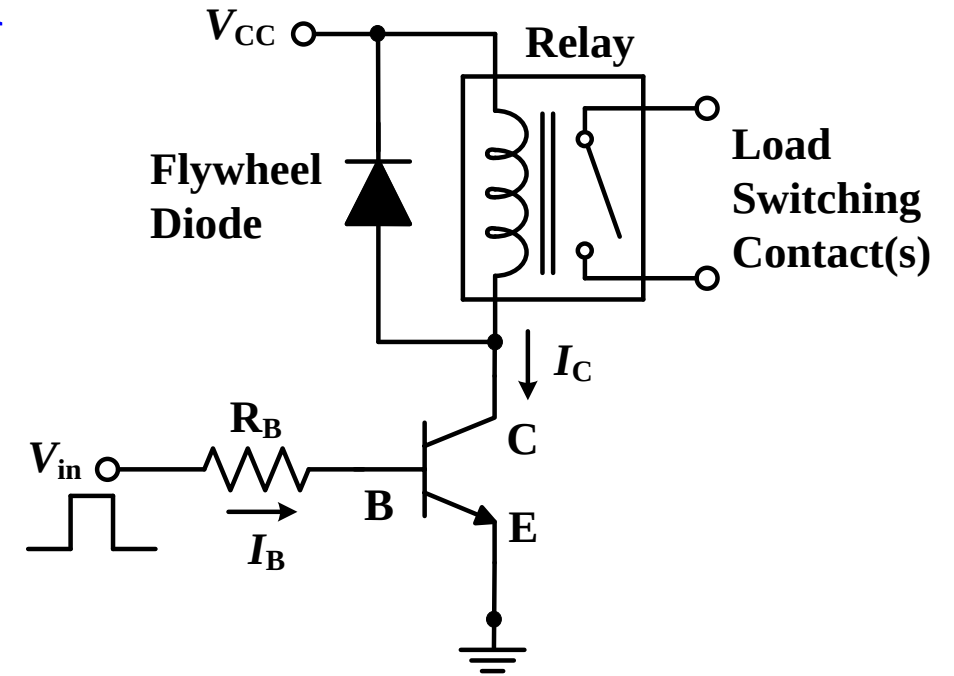
$I_L = [V_{CC} - V_{ECS} - V_{LED}] / R_2 = [5 - 0.2 - 2] / R_2 > 10\text{ mA}$
 $\Rightarrow R_2 < (2.8\text{ V}) / (10\text{ mA}) = 280\ \Omega$.
Select $R_2 = 270\ \Omega$. $\Rightarrow I_L = 10.3\text{ mA}$.
 $I_B = [V_{CC} - V_{EBS} - (V_{in})_{\text{low}}] / R_1 > I_L / \beta_{\text{min}}$
 $\Rightarrow (5 - 0.8 - 0) / R_1 > 10.3 / 50 \Rightarrow R_1 < 20.38\text{ k}\Omega$.
Select $R_1 = 18\text{ k}\Omega$.

5.6 BJT Switch for Electromagnetic Relay Control

Relay Symbols

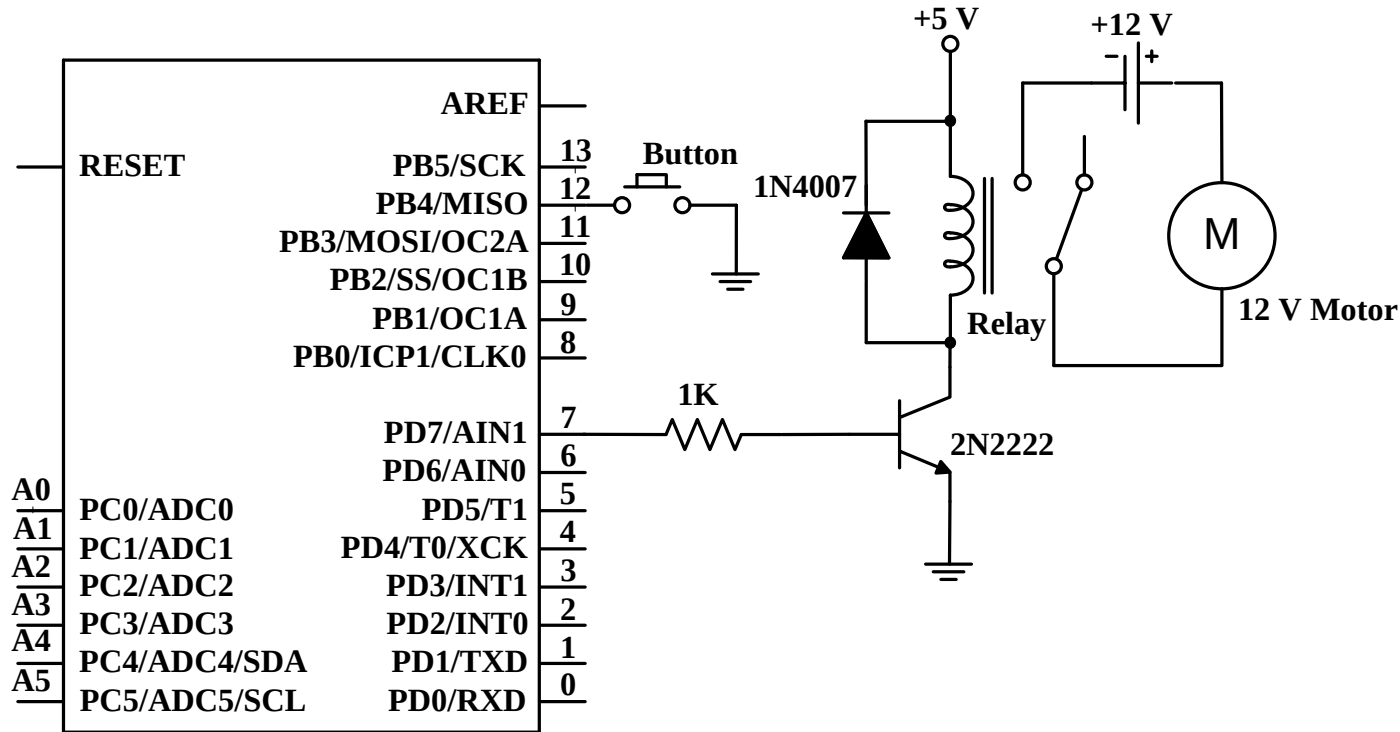


- Relay is used for switching loads with large voltage & current and for electrical isolation of the control circuit from the load voltage.
- The relay coil has a resistance and an inductance. It is activated if the coil current is above a certain threshold.
- Turning off the coil current generates a large voltage at its lower end (i.e. collector terminal). To protect the transistor from high V_{CE} , “flywheel diode” is used. When the transistor is turned off, the diode turns on, limiting V_{CE} to $V_{CC} + V_D$. The coil current slowly decays to zero.



- Relay connected to +ve supply: Use NPN transistor in CE configuration.
- The relay coil current (I_C) is switched by controlling the transistor base current (I_B).
- Transistor current gain $\beta = I_C / I_B$.
- Load switching contacts of the relay switch the load with large voltage & current.
- The relay control circuit is electrically isolated from the load.

5.7 Relay Switching Using Microcontroller (Arduino)



Example

Control of a relay with 'Arduino' digital output pin PD7 and NPN transistor 2N2222.

Relay coil current = 60 mA.

Transistor $\beta_{\min} = 30$.

Set the pin PD7 to 'Hi'. It will put the transistor in ON state, allowing current to flow through the relay coil making the relay ON.

$I_B > \text{Relay current} / \beta_{\min} = 2 \text{ mA}$.

Diode is connected in parallel with the relay coil to avoid sudden change of current in it and thereby to avoid large voltage at the transistor's collector terminal.

Source: <https://www.electronicshub.org/arduino-relay-control/>

6. Digital Logic Gates Using MOSFETs

6.1 Complementary Metal Oxide Semiconductor (CMOS) Circuits

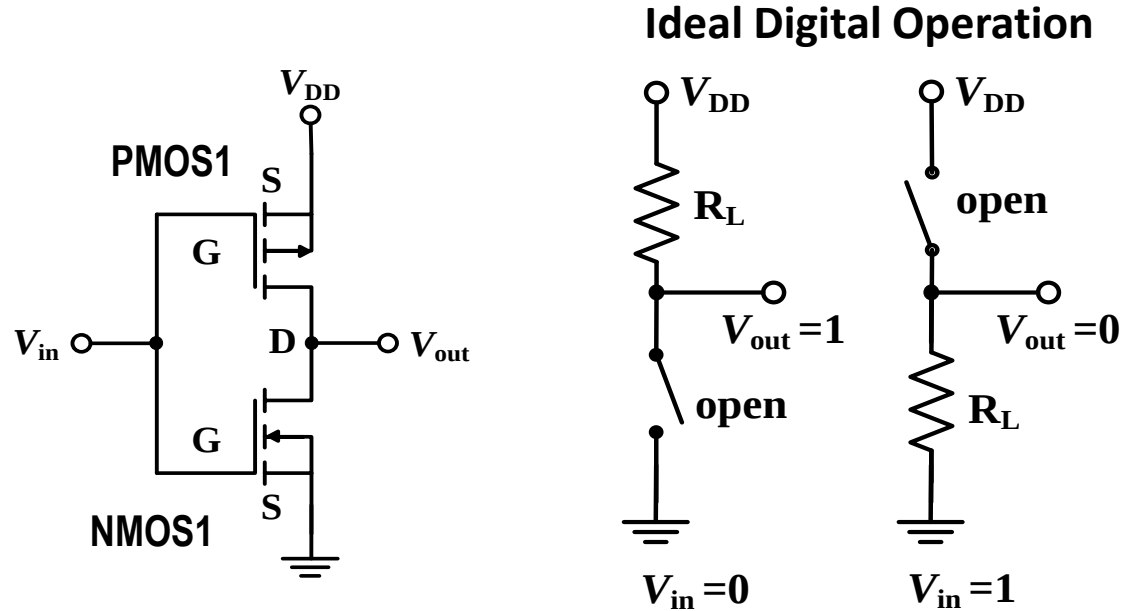
1. Combination of N-channel and P-channel MOSFETs
2. Usually same gate signal goes to N-channel and its complementary P-channel transistors for logic circuits
3. Negligible steady-state power consumption in CMOS digital circuits

6.2 Some CMOS Logic Gates

- CMOS Inverter
- CMOS NAND Gate – Universal gate
- CMOS NOR Gate – Universal Gate
- CMOS Tri-state Inverter

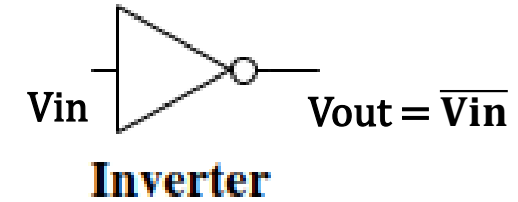
6.3 CMOS Inverter

CMOS: Complementary MOSFET, using NMOSFET & PMOSFET.

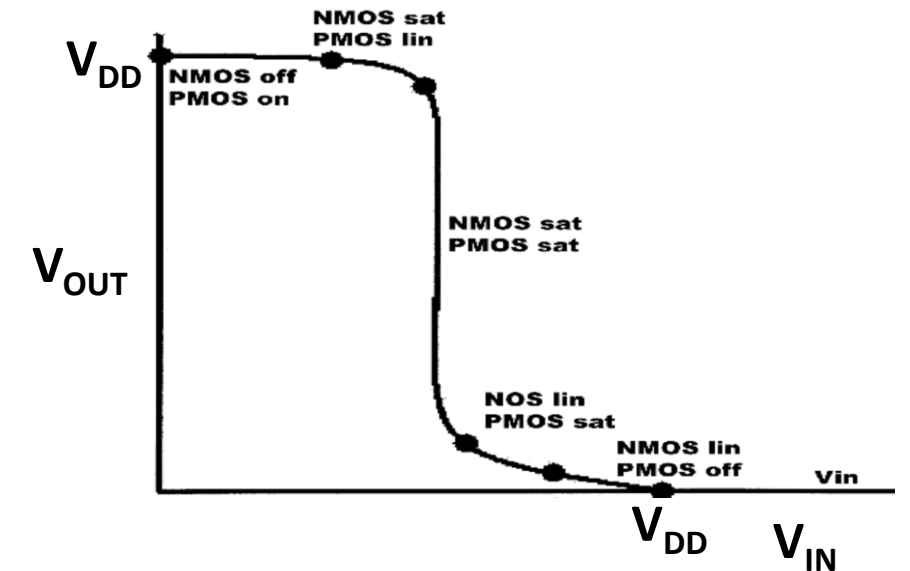


$V_{in} = \text{low (0)}$: NMOS1 is OFF (open). PMOS1 is ON (low-value R_L).
 $\Rightarrow V_{out} = \text{high (1)}$.

$V_{in} = \text{high (1)}$: NMOS1 is ON (low value R_L). PMOS1 is OFF (open).
 $V_{out} = \text{low (0)}$.



Practical Inverter Characteristics



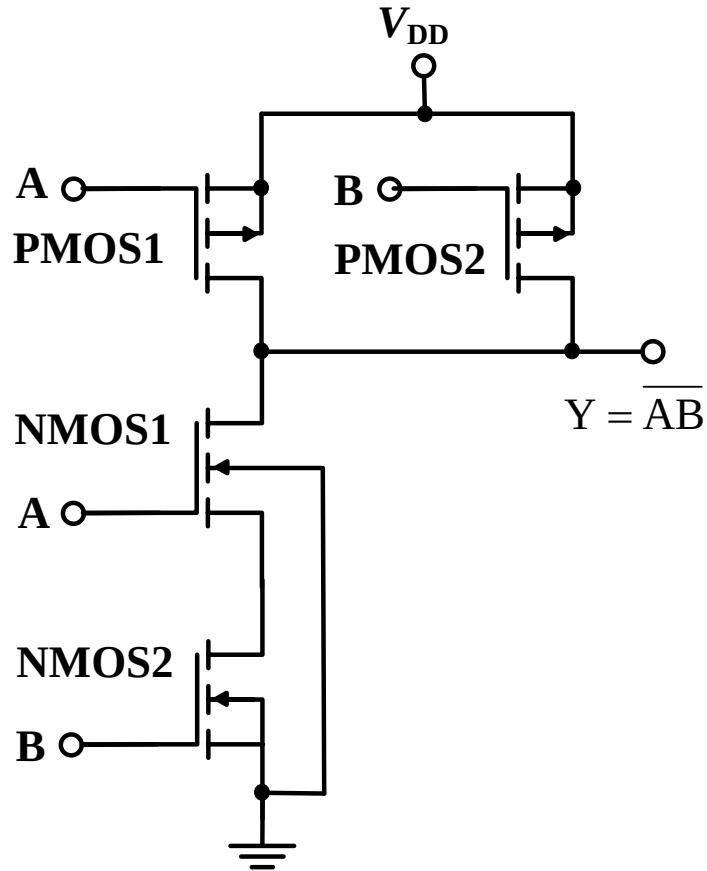
6.4 CMOS 2-Input NAND Gate

Logic 1: high voltage (V_{DD}), Logic 0: low voltage (Ground).

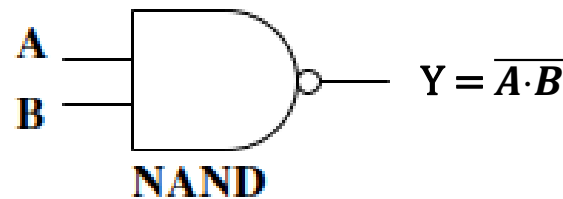
High gate voltage: NMOS is ON, PMOS is OFF. Low gate voltage: NMOS is OFF, PMOS is ON.

Y is high, if (PMOS1 ON or PMOS2 ON) and (NMOS1 OFF or NMOS2 OFF).

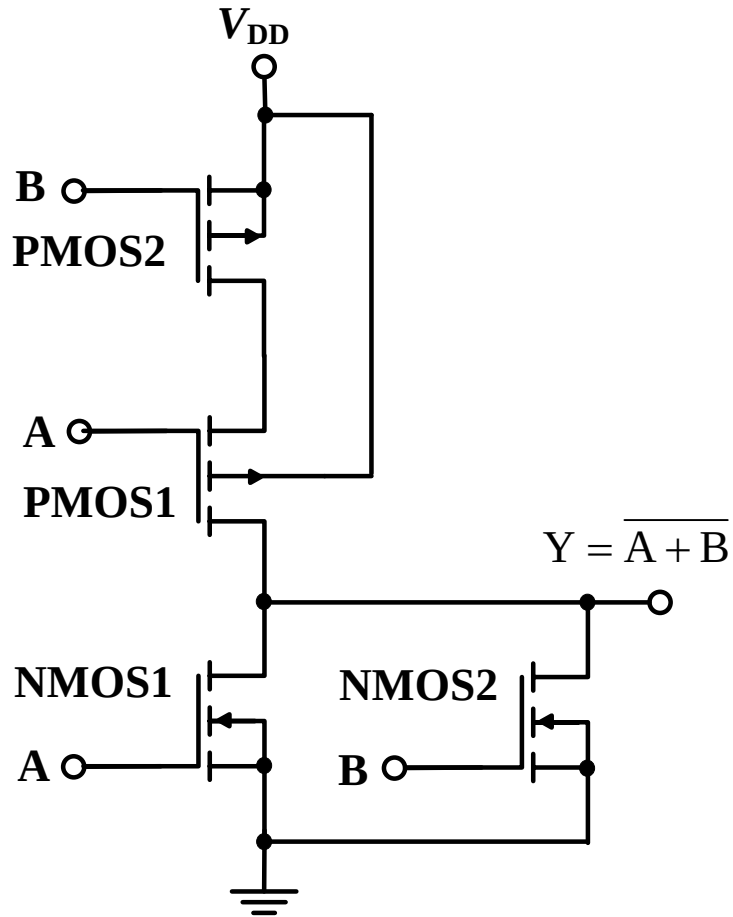
Y is low, if (PMOS1 OFF and PMOS2 OFF) and (NMOS1 and NMOS2 ON).



A	B	NMOS1	PMOS1	NMOS2	PMOS2	Y
0	0	OFF	ON	OFF	ON	1
0	1	OFF	ON	ON	OFF	1
1	0	ON	OFF	OFF	ON	1
1	1	ON	OFF	ON	OFF	0



6.5 CMOS 2-Input NOR GATE



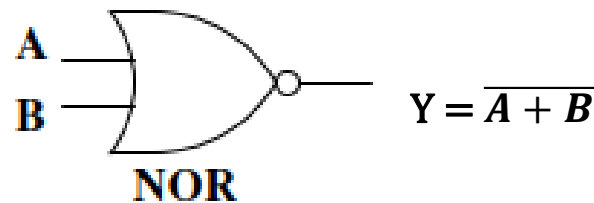
Logic 1: high voltage (V_{DD}), Logic 0: low voltage (Ground).

High gate voltage: NMOS is ON, PMOS is OFF. Low gate voltage: NMOS is OFF, PMOS is ON.

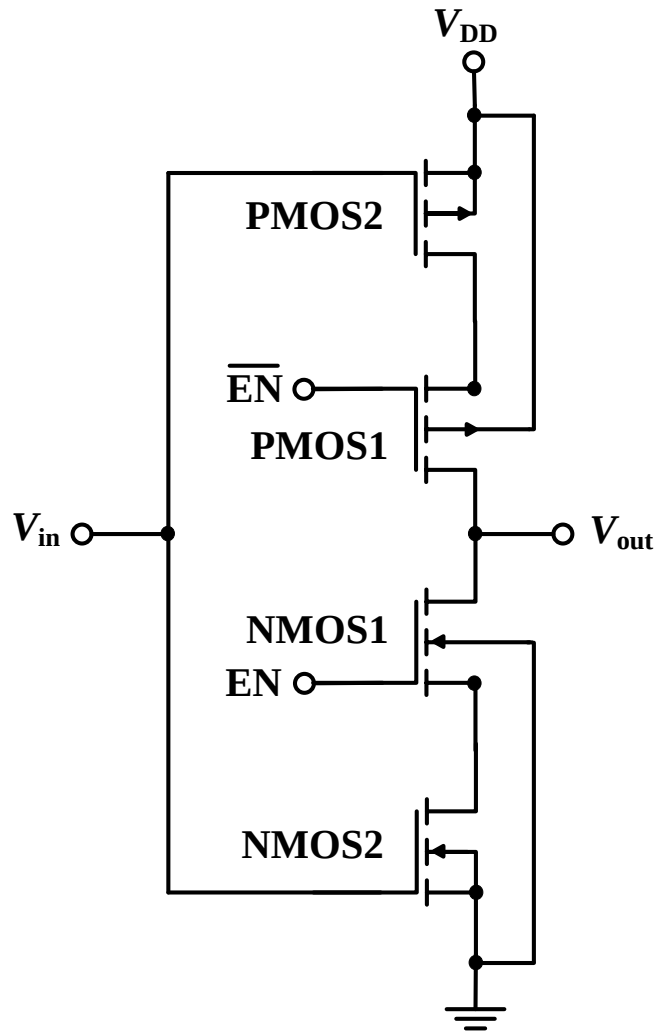
Y is high, if (PMOS1 ON and PMOS2 ON) and (NMOS1 OFF and NMOS2 OFF).

Y is low, if (PMOS1 OFF or PMOS2 OFF) and (NMOS1 or NMOS2 ON).

A	B	NMOS1	PMOS1	NMOS2	PMOS2	Y
0	0	OFF	ON	OFF	ON	1
0	1	OFF	ON	ON	OFF	0
1	0	ON	OFF	OFF	ON	0
1	1	ON	OFF	ON	OFF	0



6.6 CMOS Tri-state Inverter



$\overline{EN}$ is obtained from EN using an inverter gate.

Vout is high, if (PMOS1 ON and PMOS2 ON) and (NMOS1 OFF and NMOS2 OFF).

Vout is low, if (PMOS1 OFF or PMOS2 OFF) and (NMOS1 and NMOS2 ON).

Vout is in high impedance state, if (PMOS1 OFF or PMOS2 OFF) and (NMOS1 OFF or NMOS2 OFF).

EN	Vin	NMOS1	PMOS1	NMOS2	PMOS2	Vout
0	0	OFF	OFF	OFF	ON	High-Z
0	1	OFF	OFF	ON	OFF	High-Z
1	0	ON	ON	OFF	ON	1
1	1	ON	ON	ON	OFF	0

EN = 0 $\Rightarrow$ Vout = High impedance (High-Z)

EN = 1 $\Rightarrow$ Vout = $\overline{Vin}$

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